



Introduction to Computer Science

Unit 4: Conditionals

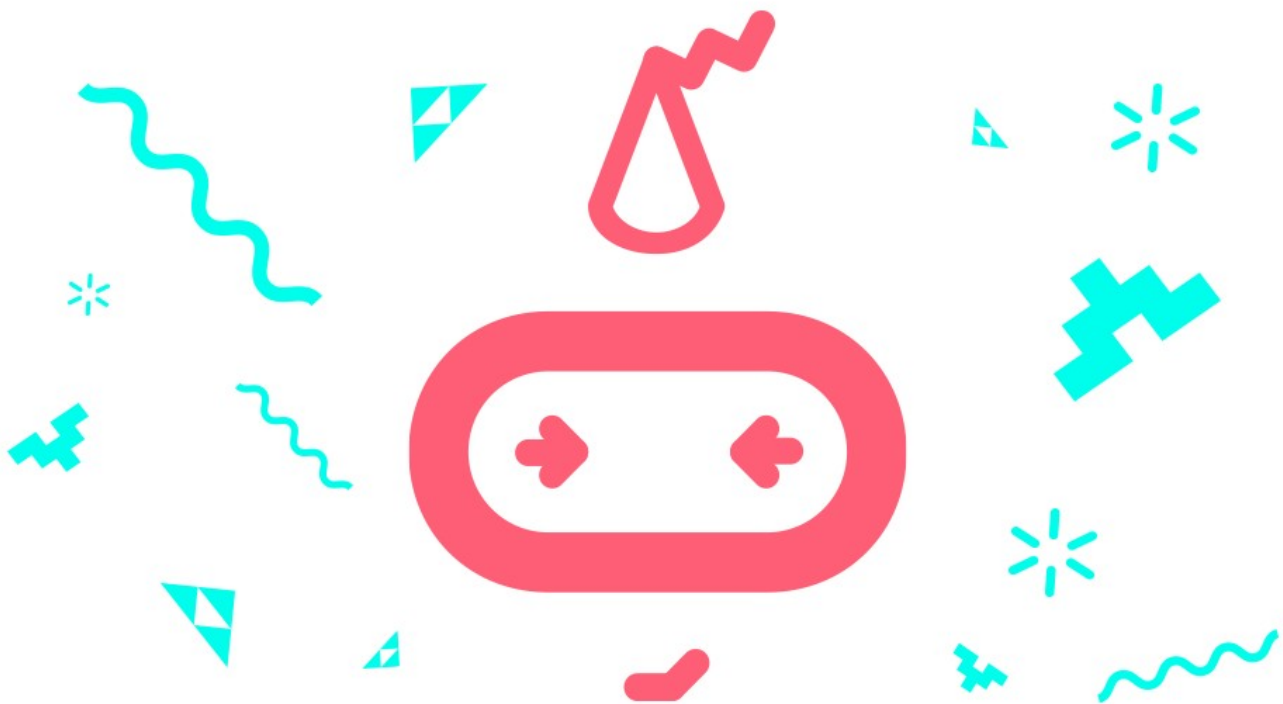
Student workbook
makecode.microbit.org



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Overview

Unit summary

This unit introduces the Logic blocks, such as 'If...then' and 'If...then...else'. You will practice skills of creativity, problem solving, and collaboration. You will learn what conditional statements are, and why and when to use them in a program. In the unplugged activity, you will experience conditionals through acting them

out in real life. Then you will code a game of “Rock, paper, scissors” with the micro:bit via the programmable buttons and the LED screen. In the final project, you will design, build, and code your own unique micro:bit-based board game using conditionals.

Learning goals

During this unit, you will:

- Understand what conditional statements are and why and when to use them in a program.
- Learn how to use the Logic blocks ‘If...then’ and ‘If...then...else’.
- Practice using the Logic blocks so different conditions yield specified outcomes.
- Demonstrate understanding and apply skills by collaborating with classmates to create a game that uses a micro:bit and a program that correctly and effectively uses conditionals.

Lesson A: Conditionals in daily life

Overview: Conditionals

Computer programs are instructions telling the computer how to process input and deliver output. An important part of programming is telling the computer WHEN to perform a certain task. For this, we use something called conditionals. *Conditionals* get their name because a certain condition or rule must be met in order for an action to be carried out.

You are all already familiar with the concept of conditionals in your daily life!

Have you ever heard your parents say:

- “If you clean your room, you can go out with your friends.”
- “If your homework is done, you can play video games.”
- “If you do your chores all week, you get your allowance, else you are grounded.”

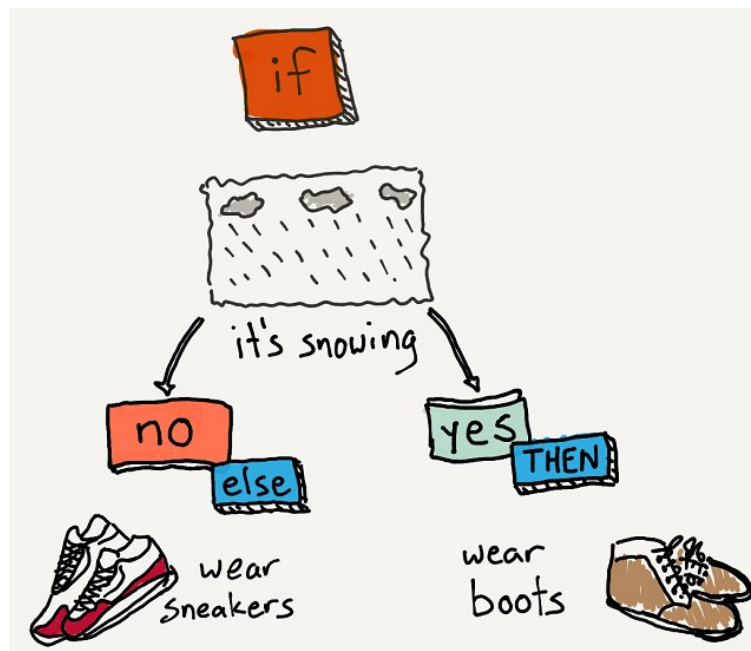
These are all conditionals! Conditionals follow the format of IF this, THEN that.

IF (condition is met), THEN (action performed)

IF (condition is met), THEN (action performed), ELSE (different action performed)

IF it is snowing, THEN wear boots, ELSE wear shoes.

The ELSE portion makes sure that a different action is performed in either case. Without the ELSE action, you might be barefoot!



Lesson B: Rock, paper, scissors

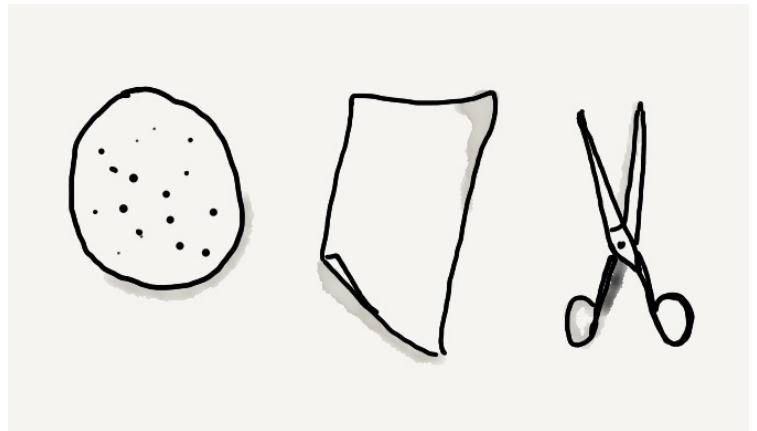
Coding activity: Rock, paper, scissors

Recall the classic “Rock, paper, scissors” game that you played in Unit 3 Variables.

- What are the rules of the game?
- What are the conditionals?

Example: If Player A gets rock and Player B gets scissors, then Player A wins.

Write the pseudocode for how to play the game on the micro:bit



Use this space to write your pseudocode

Use this space to write your pseudocode

Coding

Working from the pseudocode specifications, work in pairs to try to code a Rock, paper, scissors game on your own.

Ideas for mods

- Add a way to keep score
- Mod the game to use different images or to add more options like Rock, paper, scissors, lizard, Spock

Lesson C: Code a board game

Activity: Board game



Introduction

Many board games use an electronic toy to signal moves or provide clues. There are some funny examples online if you search for “electronic board game”. Here are some examples:

- [Dark Tower](https://youtu.be/cxrY7MWEkwE) (featuring Orson Welles) at youtu.be/cxrY7MWEkwE (0:32): This is an example of a circular board game in which the pieces start on the edges and move in toward the middle.
- [Electronic Dream Phone Board Game Commercial - 1992](https://youtu.be/pqYsQgDqlmg_) at youtu.be/pqYsQgDqlmg_ (0:30): This board game is really a logic puzzle. There are printed clues that illustrate relationships and the phone provides clues that help you narrow down possibilities by a process of elimination.
- [Stop Thief Electronic Board Game commercial -1979](https://youtu.be/q3wpPRdDy4E) at youtu.be/q3wpPRdDy4E (0:30): This board game uses a device to give audio clues that help you to figure out what to do on the game board. It's a good example of how you might use sound as a clue.

Assignment

You will work with a partner to create an original board game project in which micro:bit is a central feature, and the rules of their board game should use conditionals.

You will need to work together to come up with:

- A set of written rules (how to play)
- A game board
- A program for the micro:bit
- Photo documentation of the different game pieces, cards, or other components of the game with the micro:bit included as well as a screenshot of your micro:bit code. Each photo must have a caption that describes what the photo is documenting.
- Reflection: A diary entry describing your team's game-making process and each teammate's part in the creation of the game from brainstorming ideas through construction, programming, and testing.

The micro:bit needs to work in conjunction with the game board and/or game pieces and should be a central feature of the game. Ideally, it should be more than a simple substitute for a six-sided die.

The micro:bit might:

- Simulate the results of a battle between two pieces
- Randomly point in a different direction of travel
- Generate a result based on its current incline
- Point randomly at players and kill them
- Display a dynamic score
- Let your imaginations run wild!

Ideally, you should be writing your own versions of micro:bit programs to do something original.

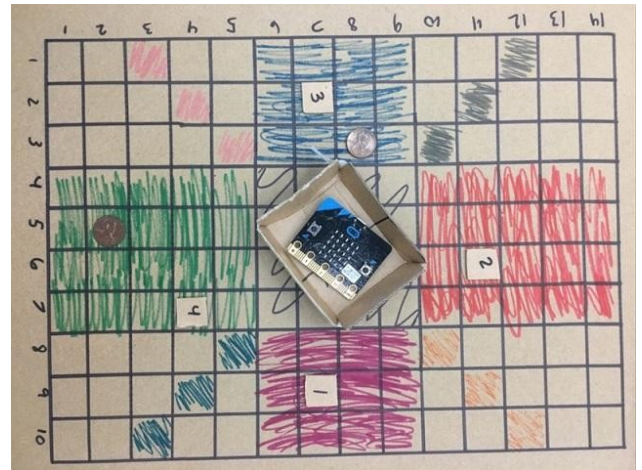
Project examples

1. Battle pieces

In this example, pieces start out at full strength and lose points based on random events on the board.

Rules: When two pieces meet on the same space, they battle.

- Press A to enter the strength of piece A.
- Then press B to enter the strength of piece B.
- Shake the micro:bit to determine the winner of the battle, which is proportionately random to the strength of each piece.



2. Space Race by K. and S.

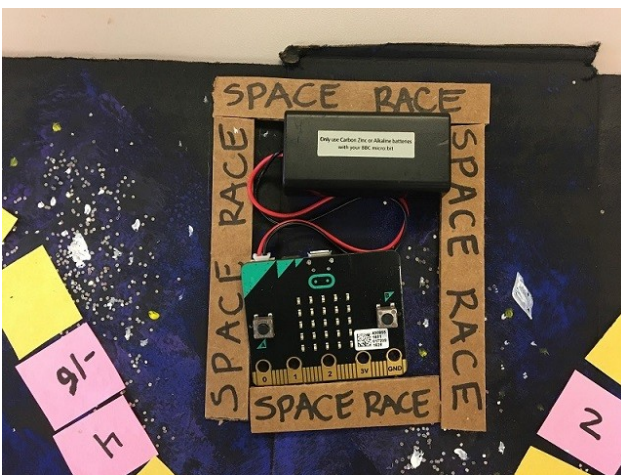
How to win: Starting from Earth, your goal is to progress to Mars. The first person to reach Mars is the winner.

Rules:

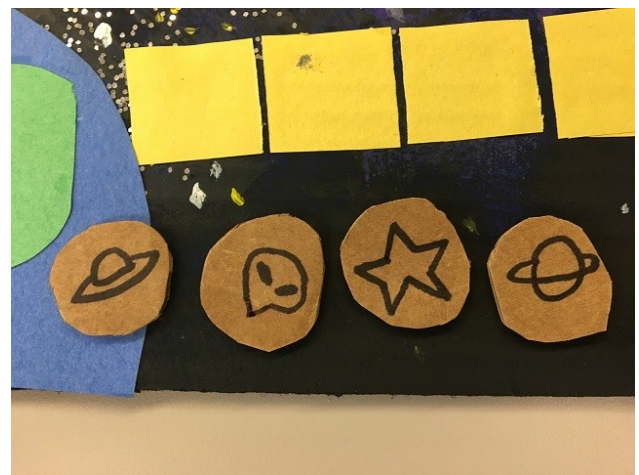
1. Shake the micro:bit to randomize how far you get to advance.
2. If you land on a pink square, press B on the micro:bit until your previous roll number appears. Then press A and B at the same time to see whether or not you move based upon the number on the square.
3. Up to four players.



micro:bit holder



Game pieces



Project expectations

Follow the design thinking approach and make sure your project meets these specifications:

- All game rules should be clear and complete.
- The game board should be complete, neat, and fit the theme of the game.
- The micro:bit must be an integral part of the game and the board.
- The program correctly uses conditionals, compiles and runs as intended, and includes comments.
- Photo documentation that includes photos of game board, code, and captions.
- Provide the written Reflection Diary entry that includes your brainstorming ideas, construction, programming, and beta testing (which we'll talk about after you complete your project).

The design thinking process:

1. Empathize by learning more about your target audience.
2. Define: Understand and identify your audience's problems or needs.
3. Ideate: Brainstorm several possible creative solutions.
4. Prototype: Construct rough drafts or sketches of your ideas.
5. Test your prototype solutions and refine until you come up with the final version.

Use this space to start your design thinking process:

Project scoring rubric

Assessment elements	1	2	3	4
Rules	Most of the game rules are missing or it is not clear what the rules are	More than one game rule is missing or not complete or not clear.	A game rule is missing or not complete or not clear.	All game rules are clear and complete.
Game board	Game board meets only one of the conditions listed for a score of 4.	Game board meets only two of the conditions listed for a score of 4.	Game board meets only three of the conditions listed for a score of 4.	Game board is: - Complete - Neat - Fits with the theme of the game - micro:bit is a central part of the game
Photo documentation	A photo is missing or of poor quality or a caption is missing.	Multiple photos and/or captions missing or of poor quality.	Most photos and/or captions are missing or of poor quality.	Complete photo documentation that includes photos of game board, code, and captions.
micro:bit program	micro:bit program lacks three or more of the required elements.	micro:bit program lacks two of the required elements.	micro:bit program lacks one of the required elements.	micro:bit program: - Uses the micro:bit in a way that is integral to the game - Uses conditionals correctly - Compiles and runs as intended - Uses meaningful comments in code

Reflection Diary

Expectations

Write a reflection of about 150–300 words addressing the following points:

- Explain how you decided, as a pair, on your particular board game idea.
- What was something that surprised you about the process of creating this game?
- Describe a difficult point in the process of designing this game and explain how you resolved it.
- What feedback did your beta testers give you? How did that help you improve your game? What were the Conditionals that you used as part of your game rules?
- Publish your MakeCode program and include the link.

Diary entry scoring rubric

Assessment elements	1	2	3	4
Diary entry	Diary entry is missing three or more of the required elements.	Diary entry is missing two of the required elements.	Diary entry is missing one of the required elements.	Diary entry addresses all elements, including: ▪ Brainstorming ideas ▪ Construction ▪ Programming ▪ Beta testing

Glossary of key terms

Conditional statement: Also known as an IF THEN or IF THEN ELSE statement. The part of a computer program or code that tells a computer when to perform an action.